



△ SHIVERING TREES

Quaking or trembling aspens are so named because their leaves, which are attached via long, flattened stalks, flutter in even a slight breeze, making the trees appear to shiver.

Pando Aspen Grove

One of the oldest organisms in the world, a quivering grove with a surprising genetic heritage



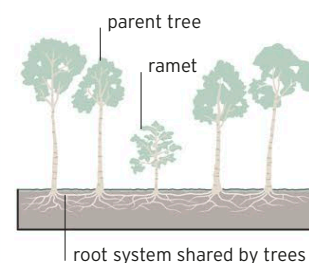
It contains what seem to be around 47,000 individual trees, but the Pando grove of central Utah is in fact one giant living organism—a clonal colony of quaking aspen. All of its trees are genetically identical, and all began as ramets, or suckers, growing from and sharing a single root system.

Ancient roots

Aspen clonal colonies are common in North America. What makes Pando—Latin for “I spread”—special is its size: it sprawls over 108 acres (44 hectares), with an estimated weight of 6,615 tons (6,000 tonnes). Pando’s accepted age is also unusual. Although individual trunks live around 100–150 years, the clonal root system is thought to be at least 80,000 years old. Some scientists, however, believe that the figure could be closer to 1 million. Recently, Pando’s survival has been threatened because deer and moose eat its ramets. Parts of the forest are now fenced off so that new suckers grow undisturbed.

SHARED ROOT SYSTEM

Aspen clonal groves begin with a single seed. As a “parent” tree grows, its lateral root system spreads. Suckers sprout from these roots, forming genetically identical trees that send out their own lateral roots, and the process continues.



The Cherokee National Forest

At the heart of an ancient mountain range, a mist-shrouded forest that is one of the most biologically diverse in North America



The Cherokee National Forest follows Tennessee’s state border with North Carolina, covering around 1,015 square miles (2,630 square km) of the Appalachian Mountains’ eastern slopes. Because the southern Appalachians were not covered by glaciers during the last ice age, diverse plant and animal species flourished, and the forests that covered the mountains provided a relatively isolated habitat until the mid-19th century. The Cherokee forest is a natural sanctuary, with more than 20,000 plant and animal species. Beech, hickory, sassafras, birch, maple, and numerous types of oaks are just some of the forest’s common trees, attracting more than 120 bird species, including the threatened red-cockaded woodpecker.



▷ MORNING FOG

The Cherokee forest is bisected by the Great Smoky Mountains National Park, and the thick, blue-tinged fogs are common to both areas.

▽ ON THE PROWL

Around 1,500 black bears roam the forest, relying on the dense cover provided by its trees.

